

DBF Final Report Comments

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To: WUDBF Contacts: skarunamoorthy@wustl.edu, mariewhite@wustl.edu,
ttuleja@wustl.edu From: DBF Head Judge

Here are comments from the judges on the Design Report your team submitted to participate in the 2019 Design/Build/Fly.

Judge Comments: This is a well written report that covers a lot of areas well. The breakdown of mission requirements into design requirements was done well, but a sensitivity study that compared the design requirements to show how they affected each other was not presented. The preliminary design section had a lot of good explanation but lacked a full set of performance estimates (had laptimes but not some other estimates such as speeds or weights). There was a prediction of lift but not drag. The detail design section was very thorough and covered all areas/components. The drawings were well done and had a lot of detail, but did not show the airplane in the stowed configuration. The testing section needed some more detail and is where more of the results should have been listed. The performance section had a lot of the testing detail so credit was given. The full aircraft performance was not detailed, such as mission times, weights, etc, and there was not a comparison to the predicted performance. The report was still very well done, overall.

Judge Comments: The biggest issue is that very little trade studies were discussed or plotted in detail. A lot of extra information was given on individual over the counter items that really did not add to the report. Many of the plots present were not well formatted (some too small to read) or explained in the text.

Judge Comments: Overall very good

Judge Comments: One general comment on this report was that it was frustrating to grade since the pdf search feature didn't work I didn't mark any points off, but it would be nice in the future if the search capability was present. The executive summary was very good, but did not mention any of the materials selected for use. The management summary schedule did not include any detail on the major sub components (e.g. wing group, tail group, fuselage). This conceptual design approach was interesting in that you effectively jumped into preliminary design for 7 configurations through the use of XFLR5 for each one. One of the goals for this section is to perform some type of sensitivity study and plot the results to show some type of optimization. While this is done to some degree with the score vs risk chart, the score assignments seem somewhat arbitrary. The preliminary design section is somewhat confusing. In 4.1 you indicate that the wing and stabilizers were sized by iterative static and dynamic stability analyses. Then in 4.3 you indicate the wing was sized from manufacturing constraints and aerodynamic analysis. In section 4.8 you are discussing wing design again. Section 4.3 indicates that a sensitivity

study was performed, indicating that score was proportional to wingspan, but the sensitivity study is never mentioned in conceptual design. Overall, it appears that there may be an over reliance on simulation tools, but little to no substantiation is provided regarding the approach. The numbers in the weights table don't sum to the total mass for mission 1. It appears to be missing 1.36 kg at -0.62 m for the M1 empty total to work out. Similarly adding 0.039 kg for the Radome appears to add 0.2 kg to the M2 total mass. The first sentence in 5.7 is an incomplete sentence. The mission performance table numbers don't seem right: Why does drag increase when the Radome is added? It also doesn't seem to make sense that the mission 1 lap time is significantly larger than missions 2 and 3. The 3-view drawing doesn't show any dimensions for the control surfaces. You also did not include a drawing of the aircraft in the stowed configuration to show that it meets the requirement. In the test plan, chart 7.1 shows initial prototype flight testing has occurred, but chart 6.2 directly above indicates that prototype integration hasn't started. The description of the test items is very light. The performance results section indicates that prototype flight testing occurred, but only provides generic details. Since the report never shows a picture of the completed aircraft (or prototype), it is assumed that it hasn't been flown as of the report submission date. The description of the flight test results does not include any comparison of the measured performance versus the predicted values.

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